

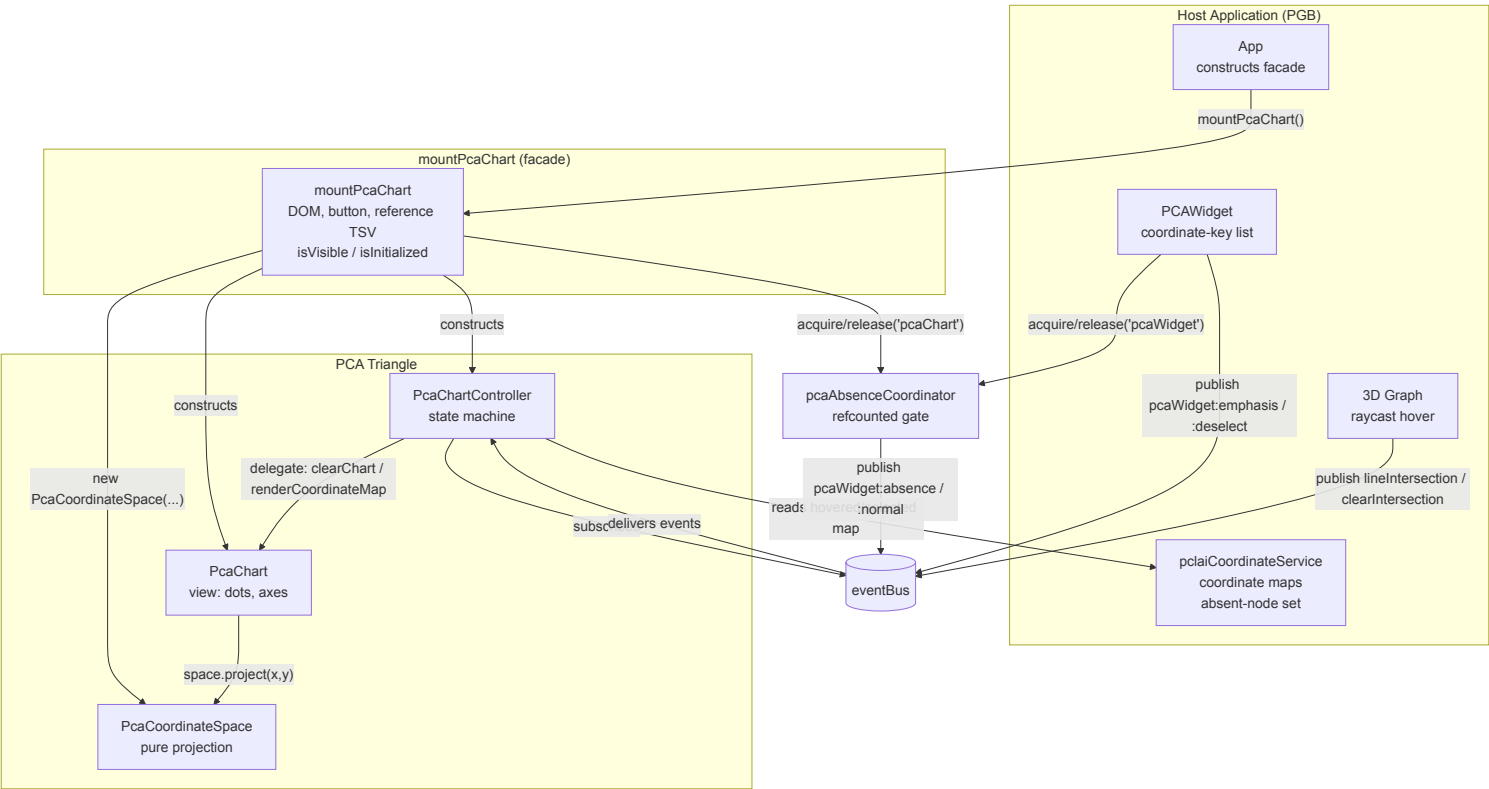
PCA Triangle Interaction Diagrams

Interaction flows for the **PCA chart subsystem** after the PR #48 refactor (closes issues #46, #47). The 970-line `pcaChartService` singleton was split into four small collaborators plus a refcount coordinator: a pure projection kernel, a view, a state-machine controller, and a bootstrap facade.

1. Architecture Overview

The PCA chart shows dataset PCLAI coordinates as dots on a 2D scatter plot with a reference-data background. It reacts to two independent signals — 3D-node hover (`lineIntersection`) and PCA-widget coordinate-key selection (`pcaWidget:emphasis`) — and renders as a pure function of `{ hoveredNodeId, selectedCoordinateKey }`.

Component	Role	Owns
App	Orchestrator	Constructs facade once in its constructor; stores handle as <code>this.pcaChart</code>
mountPcaChart (facade)	Bootstrap	Card DOM, navbar button, reference-TSV fetch, <code>isVisible</code> / <code>isInitialized</code> , absence acquire/release
PcaCoordinateSpace	Pure projection kernel	<code>project(x, y) → { left, top, size }</code> ; immutable; no DOM/events
PcaChart	View	Surface, axes, dataset dots, reference dots, hover emphasis, desaturation
PcaChartController	State machine	<code>{ hoveredNodeId, selectedCoordinateKey }</code> ; event subscriptions; single <code>render()</code> path
pcaAbsenceCoordinator	Gatekeeper	Refcounted presenter set; sole publisher of <code>pcaWidget:absence</code> / <code>pcaWidget:normal</code>
PCAWidget	Sibling presenter	Coordinate-key list card; publishes <code>pcaWidget:emphasis</code> / <code>:deselect</code> ; also acquires absence
pclaiCoordinateService	Data source	Coordinate maps, bounding box, absent-node set

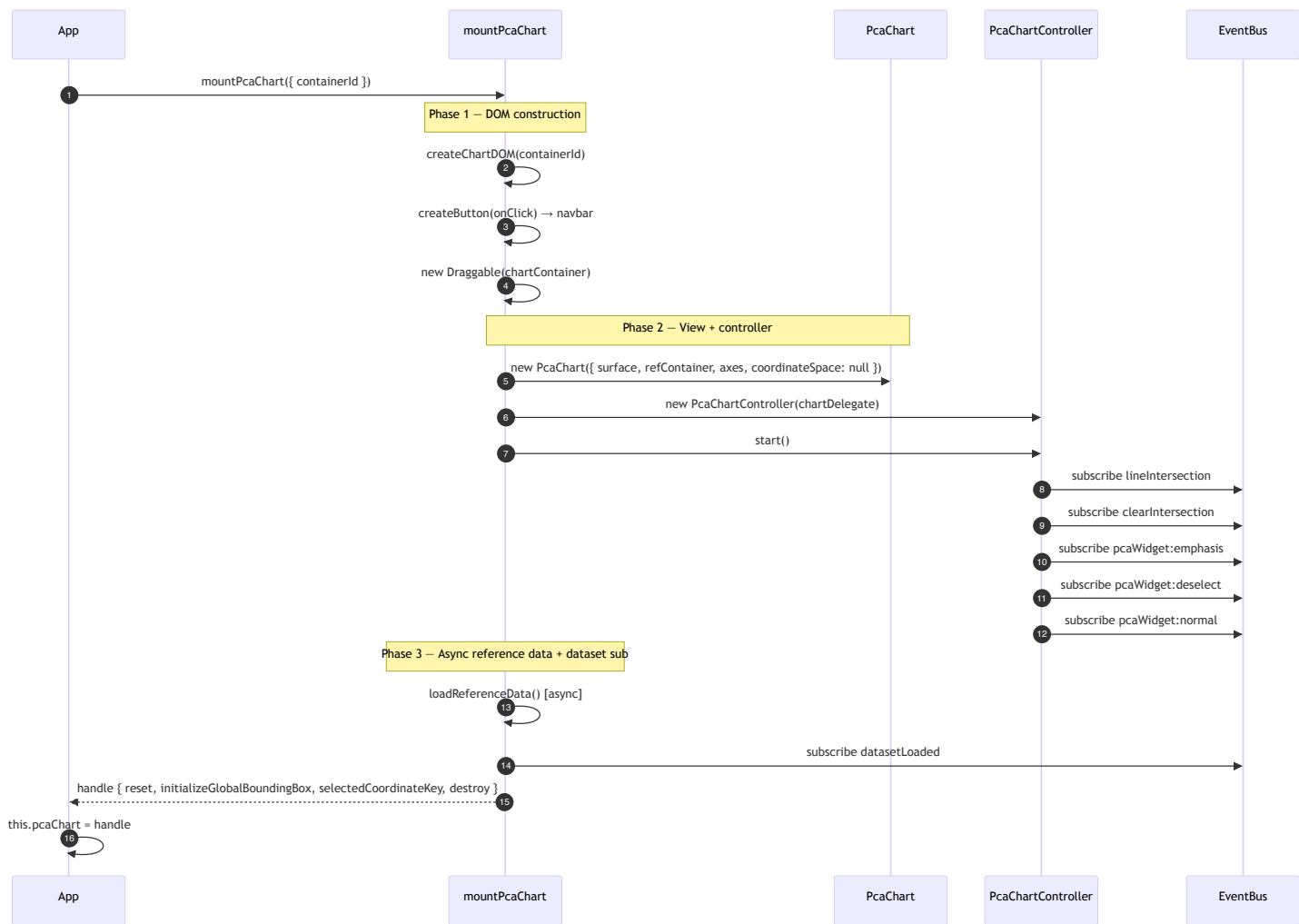


Invariants

Invariant	Where it lives
Projection math lives in exactly one place	<code>PcaCoordinateSpace</code> ; pinned by 8 characterization tests
View knows nothing about events or dataset	<code>PcaChart</code> has no <code>eventBus</code> import
Rendered chart is a pure function of <code>(hoveredNodeId, selectedCoordinateKey)</code>	<code>PcaChartController.render()</code>
<code>pcaWidget:absence / :normal</code> has exactly one publisher	<code>pcaAbsenceCoordinator</code>

2. Bootstrap — Full Sequence

`App` constructs the facade exactly once in its constructor. The facade builds DOM, wires up the view and controller, starts the reference-TSV fetch, and subscribes to `datasetLoaded` — all before any dataset has arrived.

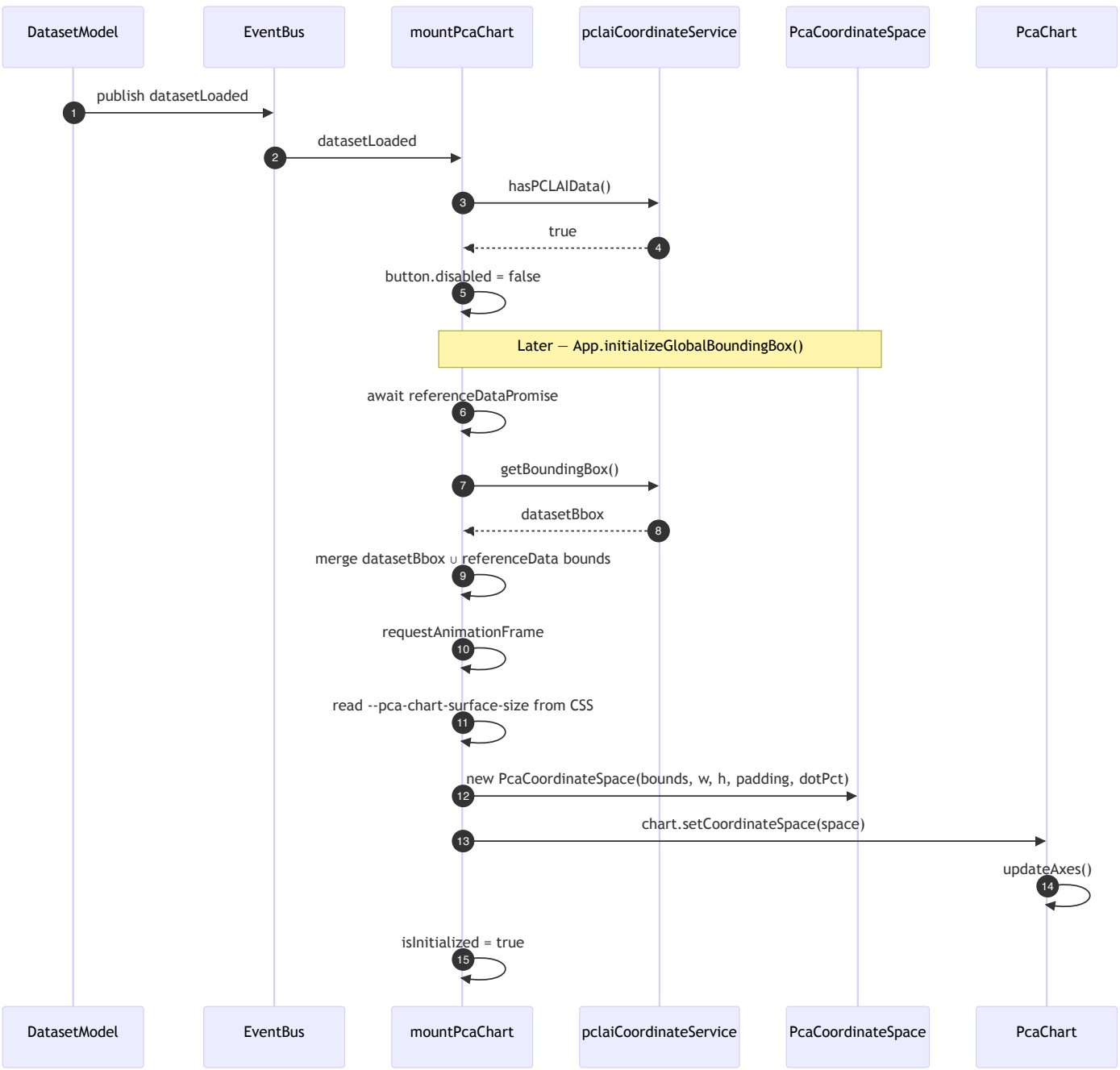


What lives where after bootstrap

State	Owner	Initial value
isVisible	facade closure	false
isInitialized	facade closure	false
coordinateSpace	facade closure	null
referenceData	facade closure	[] (filled by async fetch)
hoveredNodeId	controller	null
selectedCoordinateKey	controller	null

3. Dataset Load — Coordinate Space Construction

When a dataset loads, the facade enables the navbar button. Separately, `App.initializeGlobalBoundingBox()` awaits the reference-TSV fetch, merges dataset and reference bounds, reads the surface size from CSS, and constructs the `PcaCoordinateSpace`. Until this point, hover and widget events fire into a controller whose `isInitialized()` guard short-circuits `render()`.



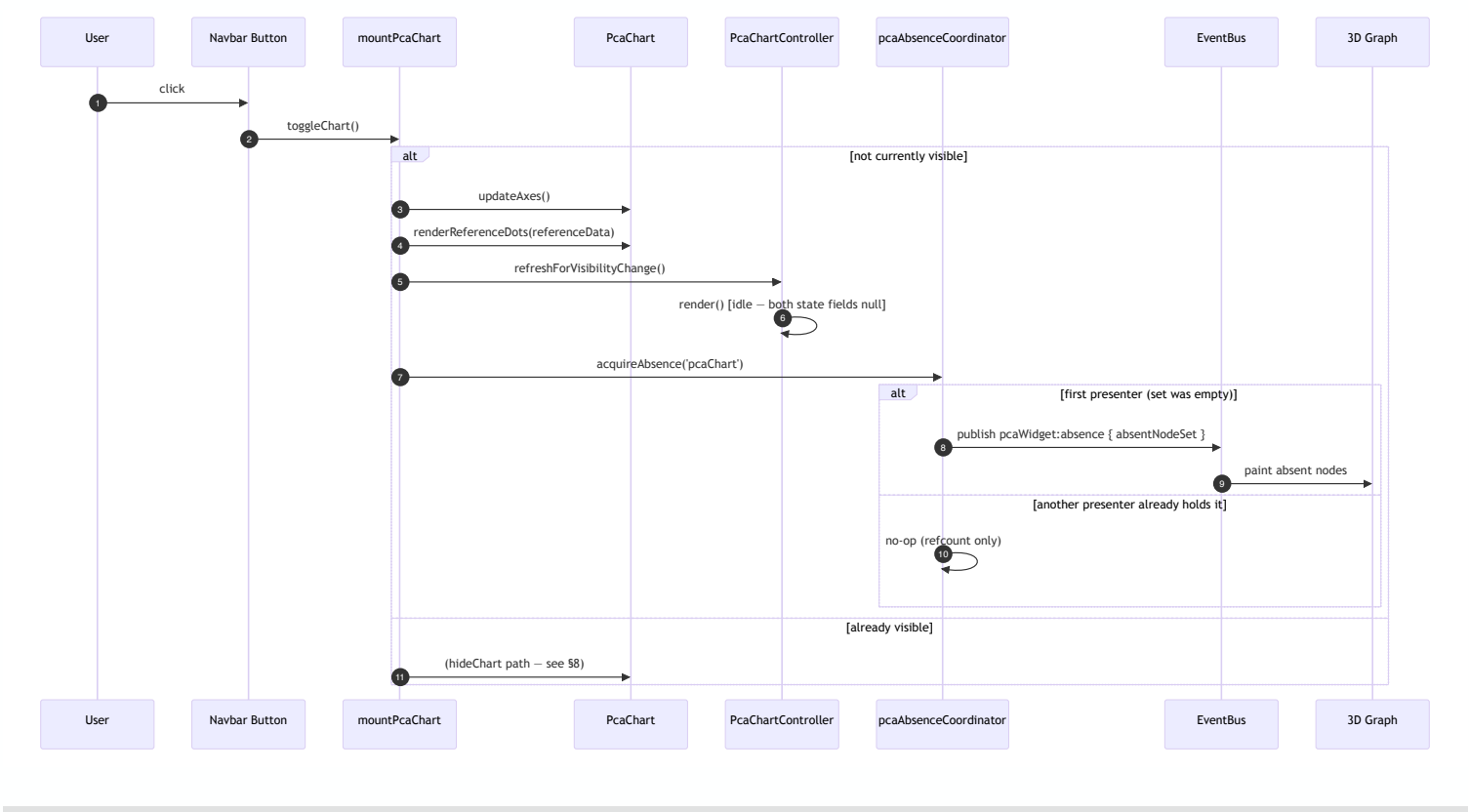
Why a two-step init

Step	Blocked on	Reason it can't happen sooner
Button enable	datasetLoaded	Needs PCLAI presence check
Coordinate	reference TSV + rAF + CSS	Bounds must union reference data; surface size is CSS-

space	read	driven
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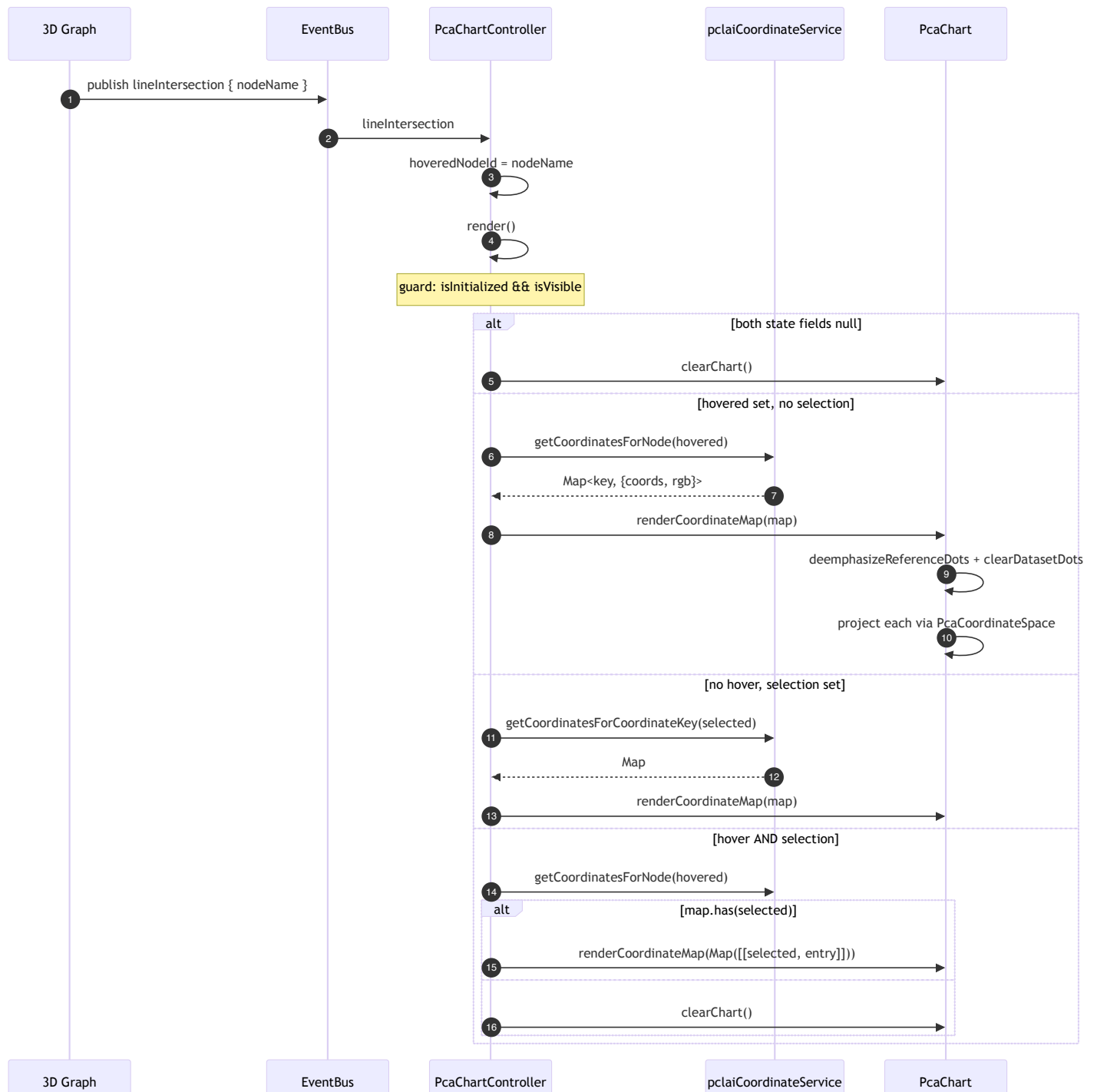
4. Show Chart — Acquire Absence

Clicking the PCA Chart button toggles visibility. Showing the chart acquires the absence presenter slot — the first presenter to do so triggers `pcaWidget:absence` , which tells the 3D graph to paint PCLAI-absent nodes in "absence" color.



5. Hover a 3D Node — Controller State Machine

The controller subscribes to `lineIntersection` and `clearIntersection` . Every handler follows the same contract: mutate `{ hoveredNodeId, selectedCoordinateKey }` , then call `render()` . The render path is a single switch on the two state fields.

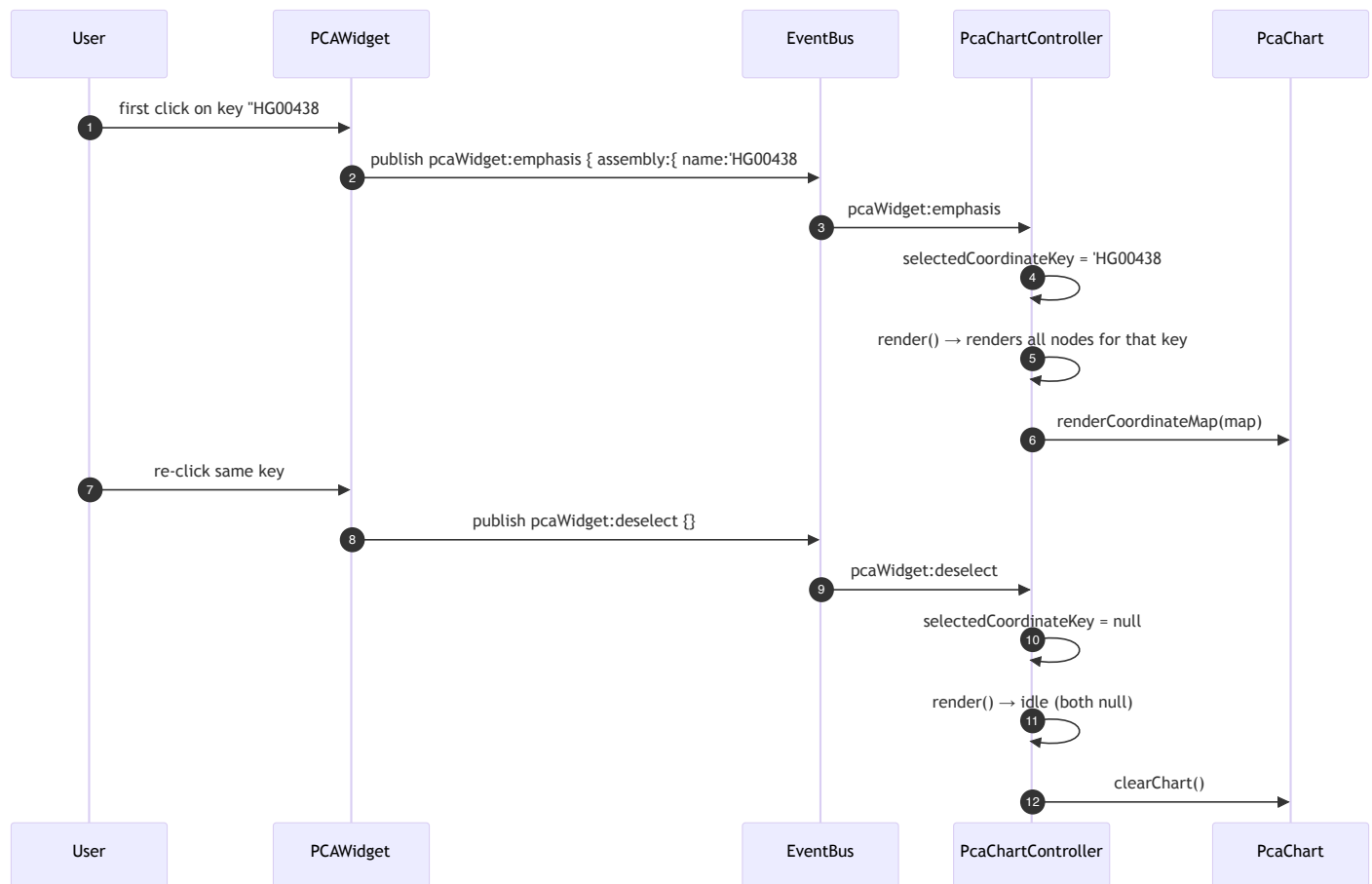


The four render branches

hoveredNodeId	selectedCoordinateKey	Result
null	null	idle — reference dots at full color, no dataset dots
set	null	all PCLAI keys for that node
null	set	all nodes that carry that key
set	set	single dot for (node, key) if present, else idle

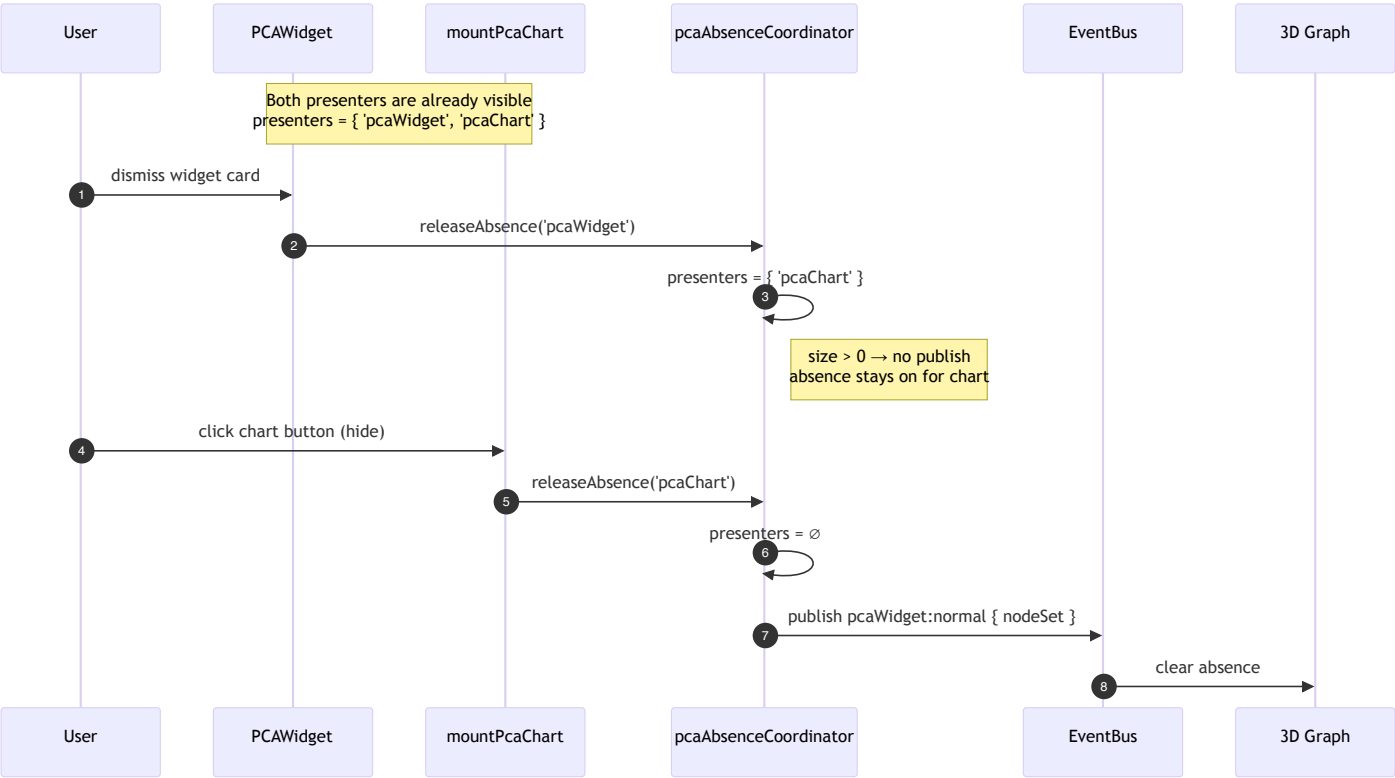
6. Widget Re-click — The #47 Fix

Issue #47 was that re-clicking the already-selected coordinate key in the widget should toggle the chart back to idle, but the old service had no channel to learn about the toggle-off. PR #48 added a `pcaWidget:deselect` event; the controller subscribes to it and clears `selectedCoordinateKey`, which falls through the pure `render()` path to idle. Once the state machine existed, the fix was two lines.



7. Refcounted Absence — Dismiss One Presenter

Both the PCA widget card and the PCA chart panel want the 3D graph in absence mode while they're visible. Without coordination, dismissing the widget would publish `pcaWidget:normal` and wipe absence state the chart still needs. The coordinator recounts presenters by ID and only publishes `:normal` when the last presenter releases.

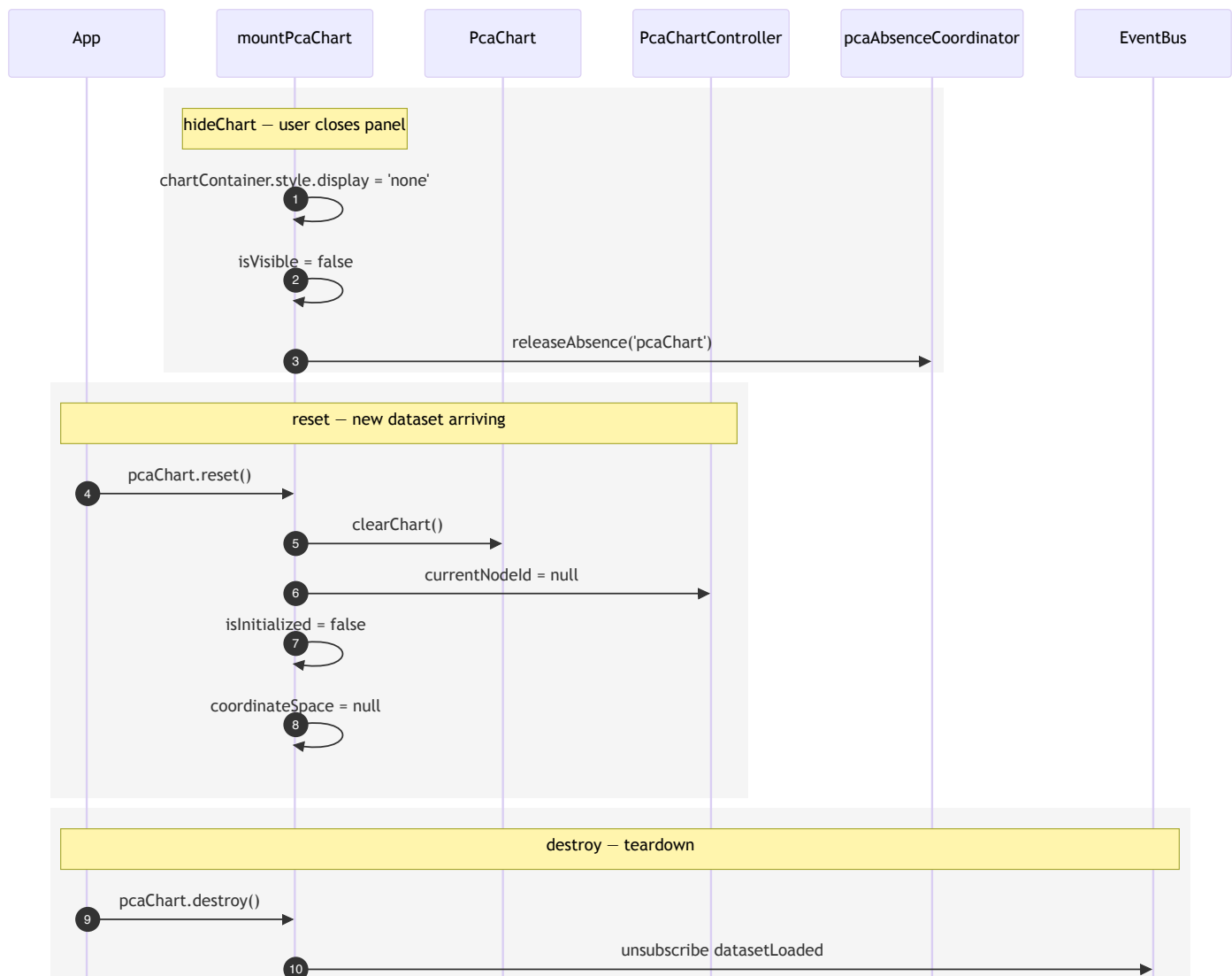


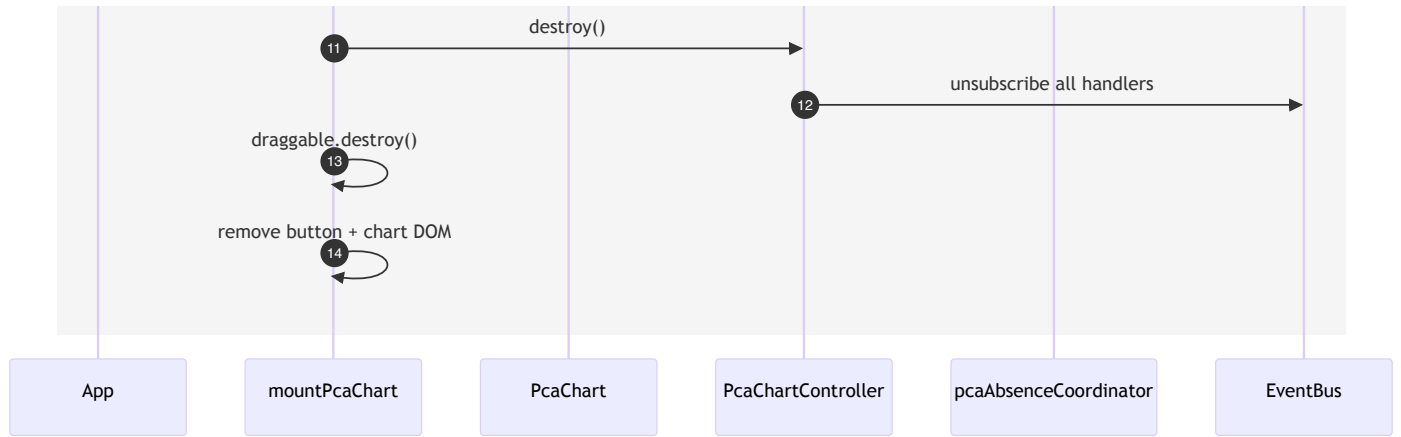
Coordinator contract

Rule	Behavior
First <code>acquire()</code> (set was empty)	Publish <code>pcaWidget:absence</code>
Re-entrant <code>acquire()</code> from same ID	No-op
<code>release()</code> from unknown ID	No-op
Last <code>release()</code> (set becomes empty)	Publish <code>pcaWidget:normal</code>

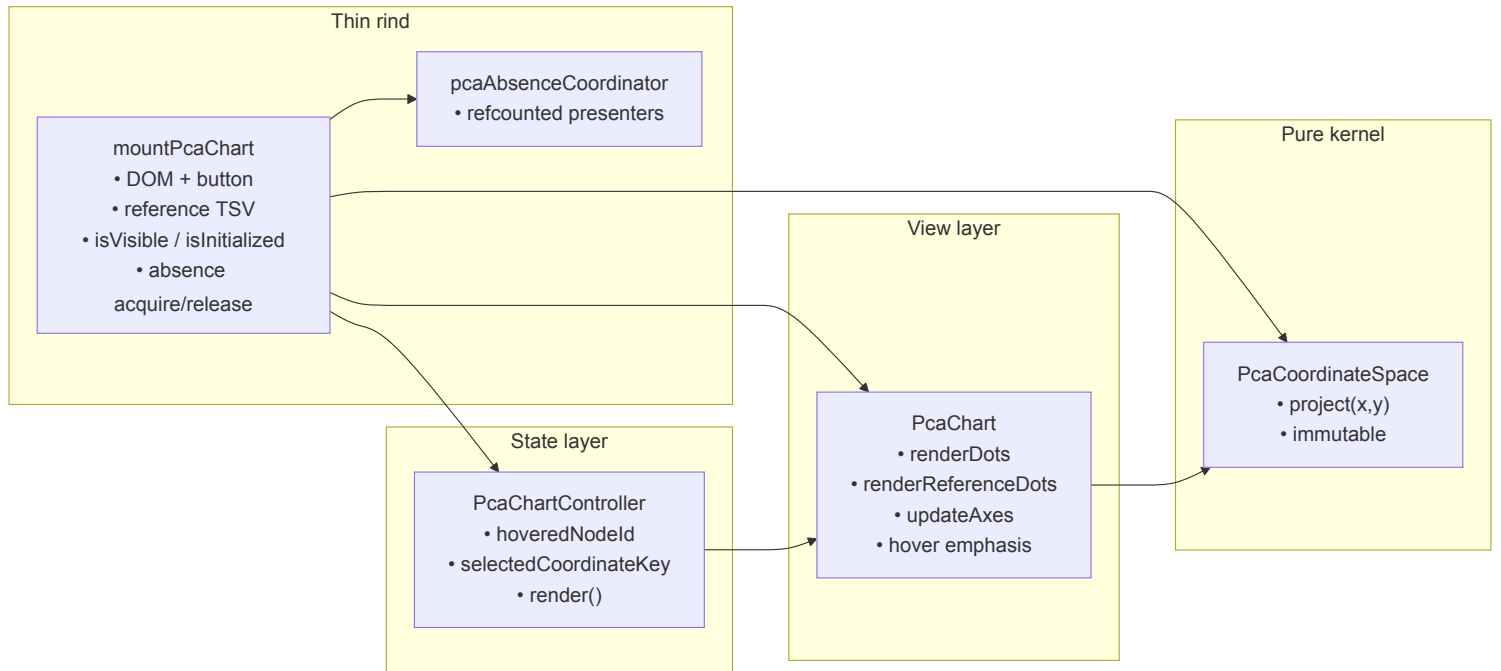
8. Hide / Reset / Destroy — Lifecycle

The facade owns lifecycle transitions. `reset()` is called on new-dataset load; `destroy()` on teardown. The controller and draggable clean up their subscriptions and DOM listeners.





9. Component Responsibilities



Module	Imports eventBus?	Imports DOM?	Imports dataset model?
PcaCoordinateSpace	no	no	no
PcaChart	no	yes	no
PcaChartController	yes	no	yes (pclaiCoordinateService)
mountPcaChart (facade)	yes	yes	yes

pcaAbsenceCoordinator	yes	no	yes
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10. Events Reference

Event	Published by	Consumed by	Meaning
datasetLoaded	DatasetModel	facade	new dataset available; update button state
lineIntersection	3D graph raycast	controller	user hovered a node line
clearIntersection	3D graph raycast	controller	hover ended
pcaWidget:emphasis	PCAWidget	controller	user selected a coordinate key
pcaWidget:deselect	PCAWidget	controller	user re-clicked the selected key (#47 fix)
pcaWidget:absence	pcaAbsenceCoordinator	3D graph Look	paint absent nodes
pcaWidget:normal	pcaAbsenceCoordinator	3D graph Look	clear absence

11. Data Flow Summary



12. Source Map

File	Role
<code>src/widgets/pcaCoordinateSpace.js</code>	Pure projection kernel
<code>src/widgets/pcaChart.js</code>	View
<code>src/widgets/pcaChartController.js</code>	State machine
<code>src/widgets/mountPcaChart.js</code>	Facade / bootstrap
<code>src/widgets/pcaAbsenceCoordinator.js</code>	RefCount gate
<code>src/widgets/pcaWidget.ts</code>	Sibling presenter (not part of the triangle, but a collaborator)
<code>src/__tests__/pcaCoordinateSpace.test.js</code>	8 characterization tests pinning the projection spec

Related: [code-architecture-improvements.md § 6](#) · PR #48 · closes #46, #47